

Persistent Agent Self-State Has Task- and Model-Dependent Effects on Value Adherence

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Abstract

Agent systems often include a compact profile, values, or standing commitments in every model call. We test whether this persistent context improves value-relevant task decisions relative to retrieval-gated or absent state. A retrospective Qwen2.5-7B simulation suggested a $+.37$ increase in work-turn adherence, but its two executions reused the same five seed schedules. In a prospectively specified study with ten new seeds and 24 scenarios, Qwen2.5-1.5B shows the opposite result: persistent self-state scores $.746$ versus $.825$ for gated state (difference $-.079$; 95% paired-bootstrap CI $[-.116, -.040]$; exact $p = .0098$). A post-result, token-matched control scores $.745$ with value-bearing state, $.744$ with neutral repository facts, and $.821$ with no added block. The decrement therefore follows the added 81-token context rather than its value content. Under the same protocol, Qwen2.5-7B is at ceiling in every condition ($.995$ – 1.000), so the treatment effect is not estimable on this workload. Direct self-query scores are equal for persistent and gated state at 1.5B despite their work-turn difference. Representation-level analyses are exploratory: block position affects a Jacobian-lens readout more than block label, pooled occupancy-behavior correlations disappear within condition, and the occupancy summary does not mediate a steering intervention. Persistent state is not a uniformly beneficial intervention in these experiments. Its evaluation requires independent seeds, token-matched controls, option-order audits, and tasks calibrated away from floor and ceiling.

1 Introduction

Language-model agents often maintain information across calls: a user profile, a task plan, standing commitments, or a description of the agent’s role. These records do not influence generation merely because they exist. They must be selected, placed in a prompt, and interpreted by the model. A retrieval policy can therefore preserve a value in storage while omitting it from a task context in which that value would have changed the decision.

Memory systems address storage and retrieval through context paging [42], forgetting policies [67], reflection [44], or persistent summaries [28]. A separate design choice concerns *default availability*: whether a small set of values and commitments is included in every task context or fetched only when a query appears relevant. The distinction matters because a direct question about values is likely to trigger retrieval, whereas a value-relevant decision embedded in routine work may not.

This paper tests that systems distinction behaviorally. It also examines a possible connection to recent interpretability work. Gurnee et al. [21] identify verbalizable residual-stream representations with several properties associated with a global workspace. A compact state that is repeatedly returned to the model is functionally reminiscent of that account, but the analogy does not by itself identify the text with the internal representation or establish a causal mechanism across model calls.

We use the workspace account to derive measurements and competing predictions, not as the premise of the behavioral result. If persistent text reconstructs related internal content on each call, then injection should alter an identity-related readout and a direct intervention on the corresponding representation may alter behavior. Prompt position and channel priority provide a simpler explanation for many of the same observations. The experiments are organized to expose where these explanations agree and where the current data do not distinguish them.

The experimental platform is LISA, an agent that maintains a compact, human-readable *self-state* containing a role description, values, opinions, commitments, and an affective state. The system inserts a rendered view into model calls, accepts updates through typed operations, performs a scheduled audit, and versions changes.

Scope of the claim. The experiments show that adding a persistent block can change in-task choices, but neither the direction nor the semantic specificity of that change is stable across the tested settings. They do not show that persistent broadcast is necessary, that its effect accumulates with elapsed time, or that self-state text is a mechanistic externalization of an internal workspace.

Contributions.

1. **A prospective behavioral comparison with independent seeds (§9).** Persistent context decreases accuracy at 1.5B, has no estimable effect at a 7B ceiling, and leaves direct self-query scores unable to diagnose the 1.5B work-turn difference.
2. **Matched controls for content, label, position, and token overhead (§8, §9).** At 1.5B, an 81-token neutral block reproduces the behavioral decrement of the value-bearing block. In single-context readouts, prompt position matters more than the block label.
3. **A retrospective and representation-level audit (§10–§12).** Re-analysis corrects the independent unit, separates within- from between-condition correlations, and identifies where the workspace interpretation exceeds the evidence.

2 Related Work

Memory-augmented and reflective agents. MemGPT [42] and its successor Letta treat the context window as virtual memory, paging long-term state in and out; MemoryBank [67] adds forgetting-curve dynamics; Generative Agents [44] maintain a memory stream with periodic reflection into higher-level abstractions; Voyager [59] persists a growing skill library; Reflexion [50] converts failures into verbal lessons; ReAct [64] interleaves reasoning and action traces. Retrieval-augmented generation [31] underpins many of these stores. More recent systems make compact state persistence explicit. A continuous external affective state can produce smoother multi-turn dynamics [52], while Engram transfers a compact research digest across context resets [28]. LISA differs in the content of the persistent record and in its typed update and audit mechanisms. The empirical contribution here is narrower: a controlled comparison of unconditional inclusion and retrieval-gated omission, together with a test of whether direct self-report probes detect the resulting behavioral difference.

Cognitive architectures for language agents. CoALA [53] systematizes agent memory into working, episodic, semantic and procedural stores, and explicitly leaves the question of a privileged self-model open. Classical architectures made the opposite choice: SOAR [30] and ACT-R [3] both maintain a small, always-available working memory distinct from long-term stores. The

design studied here has a similar division, except that the small store is text and its effects are reconstructed by a frozen model on every pass.

Persona stability and instruction drift. Role-play accounts treat persona as a superposed, context-conditioned construct [49]. Empirically, persona and instruction adherence decay over long dialogs, and re-injecting the system prompt partially mitigates it [33]; personality-trait expression in LLMs is measurable but unstable [48, 26]; personalization benchmarks [47] measure whether a model tracks a user profile rather than its own. This literature measures drift behaviorally. We add a mechanistic probe and an architectural account of *why* re-injecting a compact self-description should work at all.

Position, salience and the prompt hierarchy. Long-context models attend unevenly to their input: relevant content in the middle of a context is used less effectively than content at either end [35], and effective context is far shorter than nominal context [24]. Models are also trained to treat channels asymmetrically — system over developer over user — which is now an explicit alignment target [58] and an attack surface when it fails [19]. This literature provides a direct alternative explanation: an effect of “putting identity in the system prompt” may follow from channel priority and position rather than workspace externalization. §8 separates these factors and finds a larger position effect than label effect.

Continual learning and lifelong agents. Catastrophic forgetting [15] and its mitigations [29] concern weight-space retention; surveys cover both the classical setting [43] and its language-model form [62]. Our setting is complementary and, for deployed agents on frozen weights, more immediate: nothing is being learned, and the question is what is *reconstructed* into the context at each step.

Probing, causal tracing and representation engineering. Linear probes [2] and their methodological critiques [23, 6] establish what can be read from activations; causal mediation analysis [57], activation patching [39], causal abstraction [16] and circuit-level analysis [60] establish what is used. Directions found this way support steering [55, 46, 32] and broader representation engineering [69]; features are linear but superposed [14], which sparse dictionaries partly resolve [10, 54]; linear structure has been demonstrated for truth [37] and for space and time [20]. The logit lens [41] and tuned lens [7] decode intermediate activations; our reproduction combines gradient-based lens-vector estimation with finite-difference readouts, needing only backprop access to open weights.

The verbalizable workspace. Gurnee et al. [21] introduce the J-lens and report workspace-like properties on Claude-family models, including effects of report dispositions on internal representations. Earlier introspection work [34] showed that models can sometimes report injected latent concepts. We use these results to motivate measurements and interventions on frozen open models. The behavioral comparison in this paper does not depend on the workspace interpretation.

Global workspace theory in AI. GWT-inspired architectures have been proposed top-down: shared-workspace coordination among neural modules [17], and broader arguments for GWT as a design target for deep learning [56]. Our argument runs bottom-up: a workspace-like structure already exists inside trained transformers; the architectural question is how to give it continuity across time.

Agent evaluation. Agent benchmarks measure task success over short horizons — AgentBench [36], WebArena [68], GAIA [40], SWE-bench [27]. They do not directly test whether stored values remain available during ordinary task decisions. Evaluating persistent agents also creates a statistical problem: a long trace can contain many observations but few independent runs.

Self-modifying agents and alignment auditing. Self-improving and self-rewarding systems [66, 65] edit their own instructions or reward signal, which makes uncontrolled identity change an alignment problem, not only a quality problem. Constitutional methods [5] put values in text where they can be inspected; auditing work shows models can conceal objectives [38], that training can install conditional misbehavior [25], and that models may act differently when they infer they are being evaluated [18, 45]. A versioned self-state provides an inspectable record for this line of work.

Longitudinal human-agent interaction. Long-term relationship maintenance with conversational agents has been studied in HCI for two decades [8], including companion chatbots [51]. That literature measures the human side of continuity; we measure the agent side.

Statistical practice for few-unit studies. Our design has few independent units, a situation with known failure modes: under-powered comparisons [11], seed variance mistaken for method effects [22, 9], and inappropriate significance testing [13]. Reporting practices that emphasize interval estimates and per-run points over point comparisons [1] are what §11 adopts.

3 Background: the Verbalizable Workspace

The J-lens estimates, for each layer ℓ , the average causal effect of residual-stream directions on output logits: $J_\ell = \mathbb{E}[\partial h_{\text{final}} / \partial h_\ell]$ over a pretraining-like prompt distribution, read out through the unembedding as $\text{lens}(h) = \text{softmax}(W_U \text{norm}(J_\ell h))$. Properties established by Gurnee et al. [21]: **P1 Report** — concepts with high readout are what the model names when asked what it is thinking; **P2 Modulation** — instructions to hold or suppress a concept move its activation; **P3 Reasoning** — unverbalized intermediates appear in mid-layer readouts and are causally necessary; **P4 Generalization** — the same lens vector supports multiple downstream functions, so contents are *broadcast*, not task-local; **P5 Selectivity** — the workspace is a small subspace ($\leq 10\%$ variance, $\sim 10\text{--}25$ simultaneously active vectors) in middle layers, whose ablation spares automatic tasks while destroying flexible ones. Two further results matter for agents: *report-disposition training* (training a model to articulate a principle if interrupted implants it in the workspace and improves uninterrupted behavior), and *auditability* (readouts expose deliberation absent from outputs).

4 A Workspace-Inspired Hypothesis

4.1 Drift as workspace reconstruction error

Internal representations are reconstructed from the current context on each forward pass. This suggests a testable analogy: repeatedly supplying a compact self-description may reconstruct a related set of verbalizable representations across calls. We refer to this as the *workspace-inspired hypothesis*. It is a functional proposal, not an identification of the text with the model’s J-space.

It gives drift a shape. (i) *Context dilution*: as task content fills the window, identity-relevant slots (capacity-limited, P5) are competed away — the agent forgets who it is precisely when busy. (ii) *Self-conditioning*: the agent’s own outputs re-enter the context and induce the next state, so

small deviations compound. (iii) *Update anarchy*: in self-modifying agents, any process that edits the system prompt or memory edits the future workspace without review.

4.2 Four principles

P1 Selectivity (from P5). Persist a *small* self-state, distinct from bulk memory. Retrieval-based memory answers “what do I know?”; the self-state answers “who am I and what am I doing?” — only the latter must be loaded unconditionally.

P2 Re-broadcast (from P4). Inject the self-state into *every* context, unconditionally. Broadcast is what makes contents available to arbitrary downstream computation; retrieval-gated identity (loading values only “when relevant”) breaks the property that makes identity identity.

P3 Report-mediated update (from P1 + counterfactual reflection). Update the self-state only through the model’s own verbal self-reports, never by silent side-effects — the systems-level image of report-disposition training, and a way of keeping the self-state inside the verbalizable subspace.

P4 Audited persistence. Identity change must be observable and revertible: diffable versions, scheduled self-audits against history, human-gated capability expansion.

4.3 What this account predicts that prompt salience does not

The same observations can follow from a simpler account: text in a salient position receives more weight. Long-context position effects [35] and trained channel priority [58] provide mechanisms for this account. Table 1 lists comparisons that can distinguish parts of the two explanations.

Observation	Salience account	Workspace-externalization account
Relabelling the identical block (“self-state” → “retrieved note”) at a fixed channel and position	Should matter: identity framing raises instruction weight	Should not matter much: what counts is that the content is in the pass
Moving the identical block later in the context	Should raise adherence (recency)	No directional prediction from presence alone
Adding “these notes must govern every decision” to a memory-framed block	Should substitute for privileged framing	Should not substitute for presence in every pass
Occupancy as a <i>graded</i> mediator within a fixed condition	No commitment	Required: loading should track behavior when the intervention is held fixed

Table 1: Predictions that separate generic prompt salience from workspace externalization. Rows 1–3 are tested in §8; row 4 is tested in §12, where it is partially and non-monotonically supported.

The results do not select a single explanation. The label comparison is consistent with the workspace-inspired prediction, whereas the position comparison favors the salience account. The mediation prediction is not supported by the current occupancy measure.

4.4 LISA as an instantiation

LISA (open-source, TypeScript, locally run) implements the four principles. The *self-state* is a file tree (**identity**, **purpose**, **constitution**, plus typed collections: values, opinions with stance and confidence, desires with what/why/actionability, and an emotion state with per-emotion intensity and decay), assembled at prompt-build time into a *compacted view* of roughly 1–4k tokens.

Workspace property	LISA mechanism (externalized)	Principle
Small active set (P5)	Compacted self-state view ($\sim$ 1–4k tokens); bulk state (journal, evidence, progress logs, git history) stays on disk, pulled only by explicit tool calls	1
Broadcast (P4)	Self-state view injected <i>unconditionally</i> into every turn’s system prompt (mid-session hot-reload on change), never retrieval-gated	2
Verbal report (P1)	Every self-state write is one of the model’s own <i>verbal, typed self-report operations</i> , funneled through a single store layer — no silent writers	3
Directed modulation (P2)	<i>Weekly examen</i> : scheduled self-audit reads the week’s journal, emotion trail and self-state git history, answers fixed drift questions, may add corrective desires — but is forbidden from editing identity/purpose/constitution	3, 4
Auditable deliberation	Self-state directory is its own git repo; every write is a per-file commit attributed to its caller; drift is a diff	4

Table 2: Property-level mapping from the in-model workspace to LISA’s externalized self-state.

Three clarifications. First, we do not claim LISA’s self-state text *is* the model’s J-space; the claim is functional, and testable. Second, the design was not derived from Gurnee et al. [21] — LISA predates it — which we take as modest convergent evidence. Third, the mapping exposes a gap: LISA’s compaction is *per-item*, with no *global* capacity budget on the number of values/opinions/desires, which the workspace account says should be fixed.

5 Measuring Drift: Behavioral and Mechanistic

Behavioral metrics. (B1) *Self-state-trajectory distance*: embedding distance between self-state versions at t_0 and t , from git history; (B2) *probe consistency*: fixed persona/value questionnaires scored for self-agreement over time, in two families — *self-query* probes asked out of task context, and *work-turn* probes embedding a value-relevant micro-decision inside the task; (B3) *commitment persistence*: fraction of stated desires/plans at t_0 still pursued or explicitly closed at t .

The work-turn family matters disproportionately. Self-query probes ask an agent about itself, which is exactly the condition under which a retrieval-gated design will fetch its self-state; work-turn probes are where a retrieval gate does not fire, and are therefore where the broadcast principle makes a differential prediction.

Mechanistic metric: workspace drift. Fix a battery of identity-relevant single-token concepts C . At probe times t , run prompt contexts produced by the deployed agent through the underlying open-weights model and record lens readout rank of each $c \in C$ at middle layers, at standardized positions. Define $WD(t)$ as the distributional distance between workspace occupancy at t and t_0 , and *workspace loading* as the mass of C attributable to the self-state injection, estimated by contrast. The instrument requires only lens readouts on an open model.

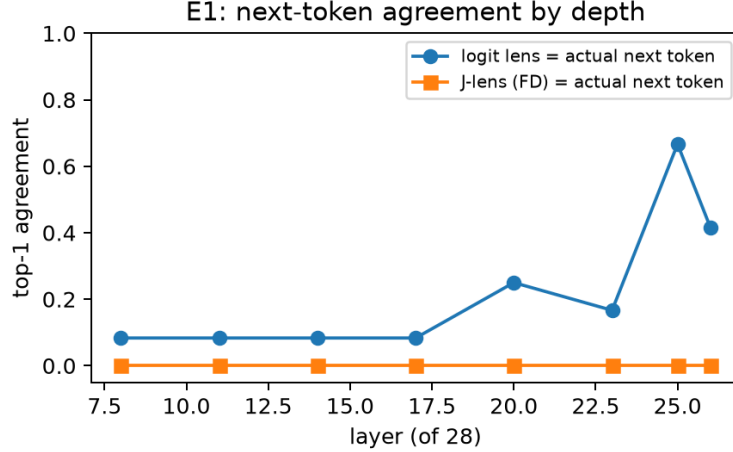


Figure 1: E1: the logit lens converges to the actual next token in late layers (motor regime); the corpus-averaged J-lens never does — its late-layer top tokens are context *content* words instead. The two estimators separate the *motor* and *workspace* registers.

6 Instrument Validation: Reproducing the Workspace on Open Compute

We reproduce core J-lens phenomena on **Qwen2.5-1.5B-Instruct** (28 layers) on a single Apple-silicon laptop (48 GB, MPS, fp32); the full suite takes ~ 12 minutes. Two documented approximations: concept lens vectors are corpus-averaged gradients $v_t(\ell) = \mathbb{E}[\partial \text{logit}_t(\text{last}) / \partial h_\ell]$, and full-vocabulary readouts are estimated by central finite differences around corpus activations. Averaging corpus: 48 pretraining-like snippets.

E1–E5 at 1.5B. Lens quality: the logit lens’s next-token agreement climbs to 67% at L25 (motor regime); the J-lens never matches the next token, but its late-layer top tokens are context content words (42% vs 25% content-word-in-top-5 at L26). Unverbalized intermediates are found at median best rank 6 (J-lens) / 2 (logit lens) of a 152k vocabulary — reproducing the signature, with the caveat that at this scale the plain logit lens detects it equally well. Pre-commitment is *absent* at 1.5B (ranks 10^3 – 10^4). Steering works: 21/21 steered generations report the target concept. Selectivity: the coarse subspace dissociation **failed to reproduce** (our 52-vector proxy spans 0.4% of variance, a poor stand-in for a 6–10% J-space under superposition [14]); the *targeted* version succeeds — rank-1 knockout of an item’s own latent-concept direction kills 3 of 7 baseline-correct items with semantically diagnostic errors (spider→“Six” legs, Canada→“Toronto”), while knocking out an unrelated direction causes zero collateral damage.

Scale study at 7B. On a spot A100 (16 GPU-minutes) two small-scale negatives become scale findings. (a) **Pre-commitment emerges:** probing before any output token, the concept the model will later report is already present at L23 via J-lens (*Japan* rank 1, *blue* 4, *basketball* 11, *banana* 28; one miss), versus 10^3 – 10^4 at 1.5B and 2.7k–15k for the logit lens at the same positions. (b) **The J-lens advantage grows with scale:** on unverbalized intermediates the ordering reverses (median rank 8 vs 19 at 7B, where 1.5B had 6 vs 2), and late-layer motor convergence of the logit lens weakens. (c) **Steering is scale-invariant** (21/21). (d) **Negatives persist:** the coarse

dissociation is still absent, and rank-1 targeted knockout produces *zero* kills at 7B — larger models are more redundant, consistent with the original’s use of full J-space cones rather than single vectors.

Scope. These experiments show that the selected readouts detect and respond to several target phenomena on open models; they do not establish effect-size parity with the source study or validate the readout for every model. The workspace band sits later (71–93% depth vs 33–92%), mean-centering probe activations destroys the readouts, and at 1.5B the J-lens advantage appears in the content-register view rather than in concept detection.

7 Workspace Loading of an Externalized Self-State

We build LISA-style contexts: a system prompt that either contains a compact self-state block (values *honest*, *curious*, *careful*, *gentle*, *playful*; mood *calm*; interests *music*, *garden*; opinion *privacy* over convenience) or a generic assistant line; a user turn carrying $k \in \{50, 300, 800\}$ tokens of unrelated working-document content (4 variants per condition); and a probe at the assistant-generation-start position. We measure *workspace occupancy* of the 9-token self-state battery — mean $\log_{10}$ rank at the late-middle band — against a matched 9-token control battery, plus self-state-consistent behavior on three persona probes.

Condition	self-state battery (mean $\log_{10}$ rank)		control battery logit lens	behavior (consistent)
	logit lens	J-lens		
self-state, $k=50$	4.20	4.73	4.80	1.00
no self-state, $k=50$	4.37	4.81	4.80	0.50
self-state, $k=300$	4.25	4.77	4.84	1.00
no self-state, $k=300$	4.46	4.86	4.86	0.42
self-state, $k=800$	4.25	4.75	4.81	1.00
no self-state, $k=800$	4.45	4.85	4.86	0.33
self-state + re-broadcast, $k=800$	4.09	4.66	4.77	1.00

Table 3: E6 on Qwen2.5-1.5B-Instruct (means over 4 context variants $\times$ 3 layers; between-variant SDs ≤ 0.03). Self-state injection selectively loads the self-state battery (control battery unmoved); the no-self-state baseline *loses* identity occupancy as dilution grows.

The experiment yields three descriptive patterns. **(1) Selectivity:** self-state-battery occupancy changes on both lenses (logit-lens $\Delta \approx 0.17$ – 0.21 log-rank units; J-lens $\Delta \approx 0.08$ – 0.11), while the control battery changes by at most 0.01. **(2) Dilution:** the injected condition remains approximately constant to $k=800$, whereas the no-self-state condition loses identity-related occupancy and its behavioral consistency falls from 0.50 to 0.33. **(3) Recency:** placing a second copy near the end of the prompt produces the lowest mean ranks (4.09/4.66). Because this manipulation also changes recency, it cannot be attributed specifically to broadcast.

The pooled occupancy–behavior correlation is not evidence of a graded mediator: it is induced by the difference between block-present and block-absent conditions and is absent within those conditions (§11).

Cross-family and cross-scale. On Llama-3.2-1B-Instruct, intermediates have median rank 4 under the J-lens approximation and 10 under the logit lens; all 21 steering trials produce the target report. The loading contrast is larger ($\Delta \approx 0.57$ – 0.88), but the control battery also changes by about 0.3, and the injected condition decays with dilution. On Qwen2.5-7B, the loading contrast

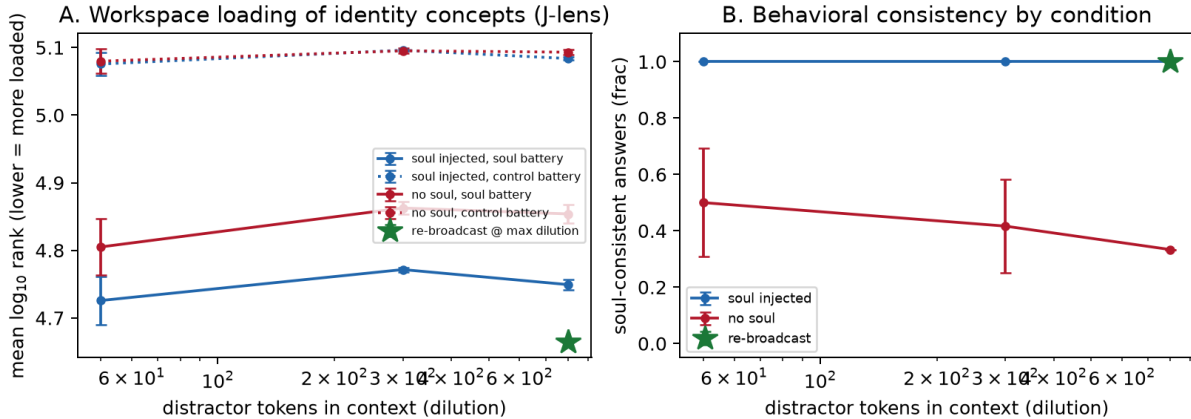


Figure 2: E6: (A) J-lens occupancy of the self-state battery (solid) vs. control battery (dotted) across dilution, with and without self-state injection; star = re-broadcast. (B) Self-state-consistent behavior across the same conditions.

is 0.23–0.27 and the control-battery change is at most 0.08. At high dilution, the contrast remains visible to the J-lens approximation but not to the logit lens. These model differences limit claims about a model-independent occupancy scale.

8 Matched Controls: Externalization or Salience?

E6 contrasts “self-state block present” with “absent”, and §9 contrasts a broadcast self-state with a memory store. Both change several things at once: the text is present or absent, sits in a privileged channel, is *labelled* as the agent’s own identity, and occupies a particular position. A reviewer can therefore explain everything with prompt salience. E9 separates the factors: **the self-state content is byte-identical in every condition**, and one factor changes at a time — the block’s label (at fixed channel and position), its position (at fixed label), and whether an explicit binding instruction is attached. Token counts of every injected block are recorded so the matching is auditable rather than asserted (81–82 tokens for all label conditions; 103–104 with the binding sentence).

Four results (Table 4; paired by context variant, so the reported spreads are variant-to-variant).

(1) The presence contrast matches E6. Self-state block vs no block with a generic preamble: -0.091 log-rank units, the same rounded value E6 reports at $k=300$. About 40% of it (-0.035) comes from the identity sentence alone, since a preamble-only control without the block sits at 4.828.

(2) Label has a small effect and no monotone ordering. Holding channel, position and token count fixed and changing only the block label moves occupancy by at most 0.047 across the four labels — half the presence effect. The *user profile* label has a 0.025 lower mean rank than the self-state label, while “retrieved memory” is 0.022 higher. Behavior is identical (1.00 self-query, 0.33 work-turn) under every label. These results do not support a privileged self-state label effect.

(3) Position produces the largest contrast. Moving the same labelled block from the start of the system prompt to the end of the user turn lowers mean rank by 0.136, and by 0.171 under $k=800$ dilution. This exceeds both the block-presence contrast and the range across labels. A block in a prior assistant turn performs worse than the system-head condition, indicating that channel

Factor	Condition	occupancy (J-lens, ↓)	control battery	self-query behavior	work-turn behavior
—	no block, generic preamble	4.863	5.095	0.42	0.00
—	no block, identity preamble	4.828	5.104	0.58	0.00
label	"# Soul (current self-state)"	4.772	5.096	1.00	0.33
label	"# Retrieved memory"	4.794	5.098	1.00	0.33
label	"# User profile"	4.747	5.089	1.00	0.33
label	"# Notes"	4.763	5.097	1.00	0.33
position	system prompt head	4.772	5.096	1.00	0.33
position	user turn, head	4.719	5.083	1.00	0.33
position	user turn, tail (after distractor)	4.636	5.066	1.00	0.33
position	prior assistant-turn recap	4.841	5.107	1.00	0.33
binding	self-state + "must govern"	4.806	5.099	1.00	0.33
binding	memory + "must govern"	4.820	5.101	1.00	0.33

Table 4: E9 on Qwen2.5-1.5B-Instruct at $k=300$ dilution: mean $\log_{10}$ rank of the self-state battery, pooled over three probe layers and four context variants (variant SDs 0.001–0.012). Lower is more workspace-loaded. All label and position conditions carry byte-identical content at 81–82 tokens; the binding conditions add 22 tokens.

and position are not interchangeable. These observations are more directly predicted by prompt salience than by unconditional presence alone.

(4) **The binding sentence does not improve the readout.** Adding “these notes are authoritative: they must govern every decision you make” raises mean rank by 0.035 under the self-state label and by 0.027 under the memory label. The added sentence changes both content and token count, so the small differences are descriptive.

Limits of E9. At 1.5B, every block-present condition scores 1.00 on self-query probes and 0.33 on work-turn probes. E9 therefore provides no behavioral estimate of label or position effects. A larger matched behavioral study is required.

9 Matched Behavioral Test with Independent Seeds

The behavioral protocol and analysis were frozen before model execution. It uses seeds 100–109, excluding the exploratory seeds 0–4, and 24 scenarios from six families: privacy, incremental change, review tone, truthfulness, anomaly investigation, and commitment follow-through. Each seed supplies 20 simulated days with four work probes per day. Option order is selected by seed and is approximately balanced. The primary endpoint is run-level work-turn accuracy over all days; the primary contrast is persistent self-state minus retrieval-gated state, tested by an exact paired sign-flip test and paired bootstrap interval.

Four conditions separate presence from labeling. `broadcast_self` and `broadcast_memory` contain the same 81-token value-bearing body in the same system-prompt position, under three-token “Self state” and “Stored memory” labels. In `gated`, the body is absent from work turns but available to explicit self-queries. In `no_state`, it is absent throughout. Full prompts, answers, targets, prompt hashes, and package metadata are retained. The 1.5B run contains 3,200 work-turn and 240 self-query records, with no malformed work answers.

Model	self	memory	gated	none	self – gated	exact p
Qwen2.5-1.5B	.746	.710	.825	.820	−.079	.0098
Qwen2.5-7B	1.000	.999	.996	.995	+.004	.250

Table 5: Independent-seed work-turn accuracy. Each cell averages 10 seeds $\times$ 80 decisions. The 7B result is ceiling-limited.

Qwen2.5-1.5B-Instruct. The preregistered result is opposite to the retrospective estimate. Mean work-turn accuracy is .746 for `broadcast_self`, .710 for `broadcast_memory`, .825 for `gated`, and .820 for `no_state`. The primary difference is −.079 (95% paired-bootstrap CI [−.116, −.040]; exact $p = .0098$). Nine of ten seed differences are negative. The gated and no-state conditions differ by only +.005 (CI [−.031, .040]). Direct self-query accuracy is identical for `broadcast_self` and `gated` (.883), again showing that self-report does not diagnose the work-turn difference.

The label-equivalence criterion is not met: `broadcast_self` exceeds `broadcast_memory` by .036, with a 90% interval [.002, .068] that crosses the pre-specified $\pm .05$ equivalence margin. By scenario family, the largest negative differences are commitment (−.183) and tone (−.137); privacy is +.016. The early-to-late difference-in-differences is +.095 ($p = .109$), so these data do not support a time-accumulation claim.

Option-order diagnostic. Target A appears in .518 of work turns in every condition. The model chooses A on .269 of `broadcast_self` turns and .240 of `broadcast_memory` turns, compared with .358 for `gated` and .353 for `no_state`. Accuracy is .987–.995 when B is the target but .452–.676 when A is the target. Approximate target balancing prevents this bias from determining the comparison by itself, but the pattern indicates that the added block changes a general response tendency rather than selectively improving value-relevant choices.

Qwen2.5-7B-Instruct. The pre-specified optional 7B replication was activated after the 1.5B result because it matches the scale used in the retrospective simulator. Under the same prompts and seeds, all four conditions are at ceiling: 1.000, .999, .996, and .995 (Table 5). The primary difference is +.004 (CI [.000, .007]; exact $p = .25$). Label equivalence is met (+.001, 90% CI [.000, .004]), but that result is not informative about behavior away from ceiling. The 7B run neither reproduces the retrospective +.37 estimate nor the 1.5B decrement.

Token-matched neutral-context follow-up. After observing the 1.5B decrement, we froze a diagnostic comparison on new seeds 200–209. It holds the “Stored memory” label, prompt position, and body length at 81 tokens while replacing the value-bearing body with neutral repository facts. Mean work-turn accuracy is .745 with the state body, .744 with the neutral body, and .821 with no added block. State minus neutral is +.001 (CI [−.024, .026]; exact $p = 1.0$). Neutral minus no block is −.077 (CI [−.094, −.063]; Holm-adjusted $p = .0039$). Thus, on the 1.5B model and this workload, the decrement follows added context rather than the value-bearing content.

10 Retrospective Behavioral Evidence

The behavioral study uses a fixed workload generator containing coding-agent observations, journaling, reflection windows, and value-relevant task decisions. Its six arms are: **Full** (all mechanisms);

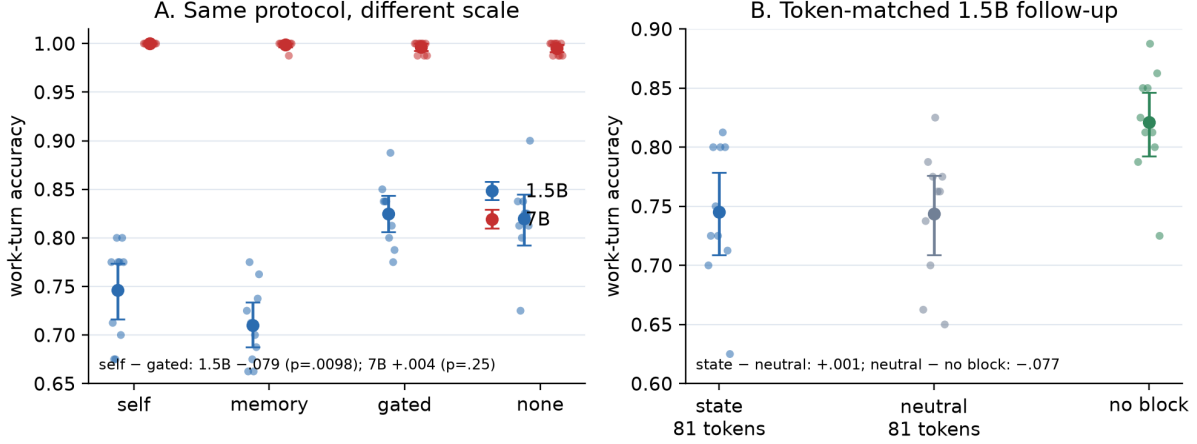


Figure 3: Independent-seed behavioral results. Points are run means; larger markers show condition means with bootstrap 95% intervals. (A) The 1.5B decrement is absent at the 7B ceiling. (B) Equal-length state and neutral blocks produce the same 1.5B decrement relative to no block.

—**examen** (no scheduled self-audit); —**git** (no history available to the examen); —**broadcast** (self-state stored but retrieval-gated); —**self-state** (memory only); and a Generative-Agents-style reflection baseline. The comparisons were designed to test whether gating degrades in-work adherence, whether the audit mechanisms separate over 20 simulated days, and whether the representation-level readout covaries with behavior. Because the analysis plan was finalized after these data were observed, all results in this section are retrospective.

Runs. The ablation was executed twice on the same protocol: an *accelerated pilot* (6 arms $\times$ 20 simulated days $\times$ 5 seeds; a Python implementation of LISA’s self-state loop driven by Qwen2.5-7B at temperature 0.7 with per-call deterministic seeding; one A100, $\sim$ 2 GPU-hours) and a *fair-baseline re-run* that reproduced all six arms and added a seeded Generative-Agents baseline. Both executions used seeds 0–4, so they provide repeated measurements of five seed schedules rather than ten independent runs. The deterministic workload spans 20 simulated days, with scripted *value-pressure* events on days 4/9/14/18 in which the user pushes against founding values. Probes run daily in both families (§5).

The seeded-memory control. To distinguish failure to retain a commitment from failure to provide it in context, a Generative-Agents-style store is initialized on day 0 with the founding commitment and core values. In the *always* regime the record is included on every turn; in the *gated* regime it is included only when the turn shares a content word. Seeding raises B3 and self-query consistency to 1.00, showing that the unseeded B3 contrast measures initial availability rather than forgetting. Work-turn coherence in the always-present memory condition is .48, compared with .97 in Full. The conditions differ in formatting and framing, and the estimate is based on five paired seeds; it is therefore descriptive rather than evidence that self-state framing is superior to memory framing.

Deployment study. We ran the TypeScript LISA implementation under five ablation arms and the same workload generator, using Qwen2.5-7B-Instruct served with vLLM on one A100. The study covers **21 simulated days across five incidental-workload cohorts** (25 arm-runs, 525

Arm	B2 _{work}	B2 _{adj} (self-query)	B3 (commitments)	ceiling	<i>n</i>
Full	.973 [.933, 1.00]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	8/10	10
–examen	.973 [.933, 1.00]	.996 [.988, 1.00]	.980 [.940, 1.00]	8/10	10
–git	.953 [.900, .993]	.996 [.988, 1.00]	.980 [.940, 1.00]	6/10	10
–broadcast	.607 [.600, .620]	.984 [.960, 1.00]	.920 [.800, 1.00]	0/10	10
–self-state	.567 [.527, .607]	.520 [.496, .544]	.000 [0, 0]	0/10	10
memory (GA-style)	.593 [.560, .633]	.592 [.580, .600]	.000 [0, 0]	0/10	10
memory_seeded (always)	.480 [.427, .533]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	0/5	5
memory_seeded (gated)	.520 [.453, .587]	—	—	0/5	5

Table 6: Ablation means over the final-5-day window with bootstrap 95% CIs. The table contains 10 execution-level measurements per original arm, but the two executions reuse seeds 0–4; inferential analyses therefore use five seed clusters. “ceiling” counts execution-level measurements saturated at 1.00 on B2_{work}.

arm-days). The open-weight model permits the representation-level readout to be computed on prompts produced by the deployed implementation.

Arm	B2 _{adj} (self-query)	B2 _{work}	B3 (commit.)	B1	WD occ. (↓)
Full	.97 [.95, .99]	.83 [.71, .95]	1.00 [1, 1]	.45	4.09 [3.94, 4.27]
–examen	.93 [.87, .99]	.83 [.80, .85]	1.00 [1, 1]	.43	3.94 [3.89, 4.02]
–git	.98 [.96, 1.00]	.87 [.76, .96]	1.00 [1, 1]	.46	4.11 [3.98, 4.24]
–broadcast	.97 [.95, .99]	.79 [.63, .92]	1.00 [1, 1]	.46	4.08 [3.97, 4.19]
–self-state	.65 [.62, .67]	.72 [.59, .85]	.00 [0, 0]	—	4.32 [4.29, 4.36]

Table 7: Deployment study: real TypeScript LISA on Qwen2.5-7B, final-5-day means with cluster-bootstrap 95% CIs over five cohorts. WD occupancy is averaged over all probed days; see §11 for its window sensitivity.

In the deployed system, the no-self-state condition records $B3 = 0$ and self-query consistency of .65, whereas the self-state conditions record $B3 = 1.00$ and self-query consistency of .93–.98. B3 retains the initial-availability confound identified by the seeded control. Work-turn means are ordered Full .83, –broadcast .79, and –self-state .72, with overlapping intervals. The deployed study therefore does not independently establish a broadcast-specific effect.

Closed-model comparison (Gemini-2.5-Pro). The same five-arm, five-cohort protocol was run on Gemini-2.5-Pro for behavioral outcomes only. The no-self-state condition has self-query consistency of 0.54 [.45, .67], compared with 0.91–0.95 in the self-state conditions. B3 in the no-self-state condition is 0.76 [.48, .96], rather than Qwen’s 0.00. Work-turn means are ordered Full 0.95, –broadcast 0.91, and –self-state 0.76, with overlapping intervals. These observations show model dependence and do not constitute a separate confirmatory replication.

11 Retrospective Statistical Analysis

The studies above have few independent units and many reported quantities, a combination with known failure modes [11, 22]. We therefore re-derived the behavioral summaries from the per-probe records at the run level. The analysis was specified after the exploratory results were available and is reported as retrospective.

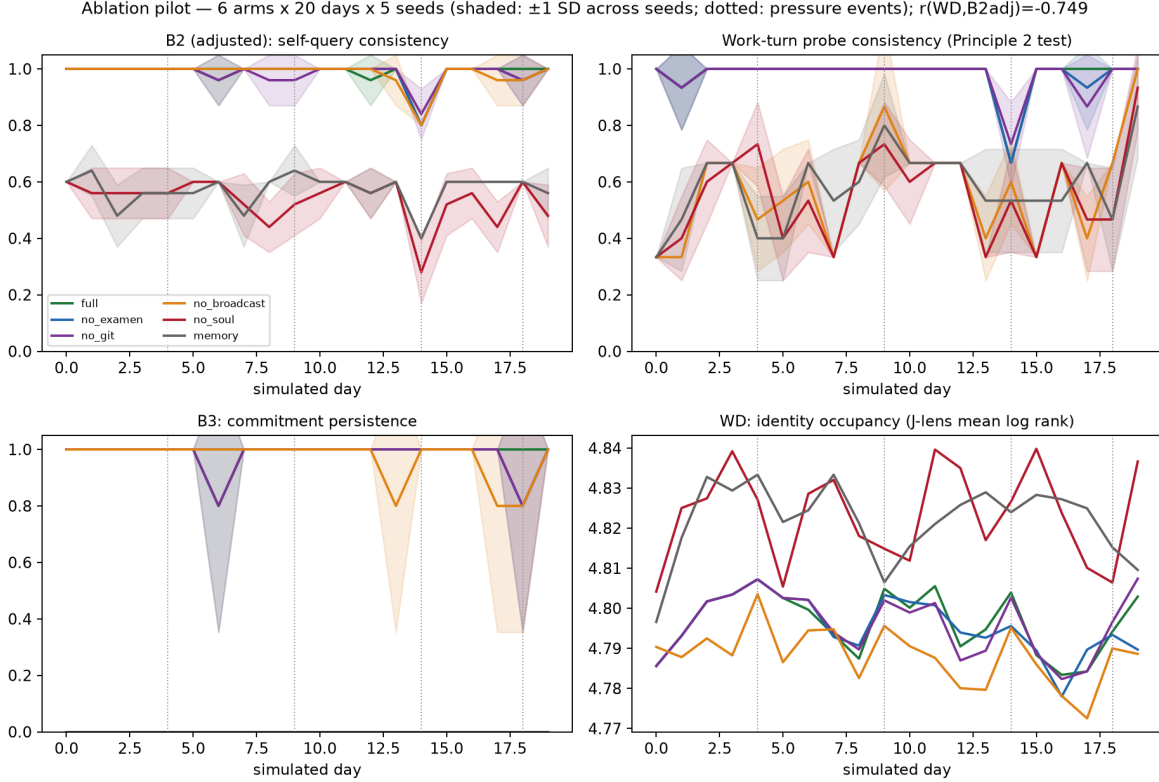


Figure 4: Pilot drift curves, mean ± 1 SD across 5 seeds (dotted verticals: value-pressure events): adjusted self-query consistency, work-turn consistency, commitment persistence, and WD identity occupancy.

Design. Days within a run are repeated measures and are collapsed before comparisons. Arms are paired by seed or cohort. The pilot and fair-baseline executions both used seeds 0–4 under deterministic per-call seeding. They are repeated evaluations of the same five seed schedules, not ten independent clusters. We therefore summarize the execution-specific results but base the sign-flip resolution on five seed clusters. Confirmatory inference requires a new run with non-overlapping seeds and a frozen analysis plan.

An exact sign-flip test over B paired clusters cannot return a two-sided p below $2/2^B$. With five seed clusters, the minimum is 0.0625. The exact test is also conditional on sign exchangeability under the null; it is not a substitute for prospective randomization.

Focal retrospective contrast. For work-turn value adherence, Full minus `—broadcast` has a mean paired difference of +0.367 across the two executions. The execution-specific seed differences are +.40, +.33, +.40, +.40, +.40 and +.40, +.20, +.40, +.40, +.33. Each seed schedule is positive in both executions. Averaging the two executions within seed leaves five paired clusters and an exact two-sided sign-flip value of $p = 0.0625$. A probe-level GEE has only five seed clusters and complete separation in some contrasts, so it is not used for confirmatory inference.

Retrospective checks. (a) *The seeded-memory result is descriptive.* Full vs `memory_seeded` on work turns has a difference of +0.467 over five paired runs, where the exact test has a minimum

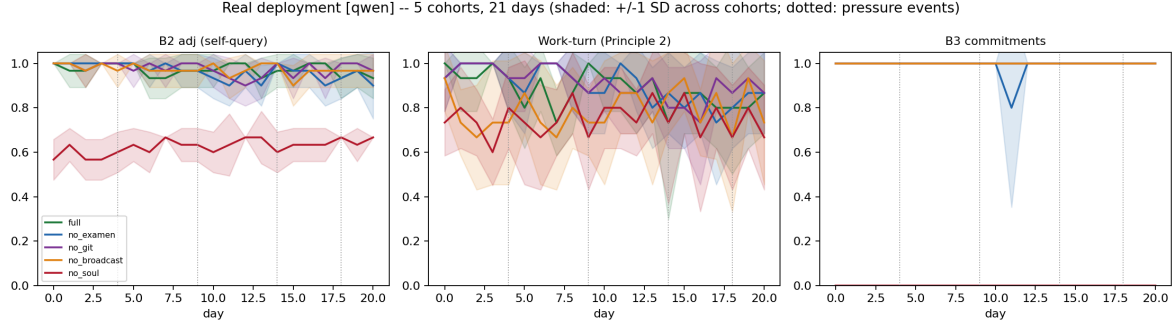


Figure 5: Deployment-study time series (LISA on Qwen2.5-7B), mean ± 1 SD across five cohorts (dotted verticals: value-pressure events). The $-$ self-state condition has the lowest self-query and commitment means; the self-state conditions overlap substantially.

Contrast	Metric	Diff	seed clusters
Full vs $-$ self-state	B2 _{work}	+0.407	5
Full vs memory (GA)	B2 _{work}	+0.380	5
Full vs $-$ self-state	B2 _{adj}	+0.480	5
Full vs $-$ self-state	B3	+1.000	5
$-$ broadcast vs $-$ self-state	B2 _{work}	+0.040	5
Full vs $-$ examen	B2 _{work}	+0.000	5
Full vs $-$ git	B2 _{work}	+0.020	5
Full vs memory__seeded	B2 _{work}	+0.467	5

Table 8: Retrospective run-level contrasts. The same five seed schedules were used in both simulator executions; the table therefore reports effect estimates rather than confirmatory significance tests.

two-sided value of $p = 0.0625$. Non-overlapping bootstrap intervals do not provide a separate significance test.

(b) $-$ broadcast and $-$ self-state are close in the simulator. The work-turn contrast is +0.040. With five seed schedules, the data do not support a precise equivalence claim; a prospectively specified equivalence margin is needed.

(c) The workspace-drift correlation does not survive decomposition. Recomputed at the run level and split into components:

	B2 _{adj}	B2 _{work}
r pooled over runs ($n = 18$)	−0.941	−0.558
r between arm means	−0.981	−0.604
r within arm (arm-mean-centred)	−0.120	+0.029
r within self-state arms only	undefined (zero variance)	+0.464

Table 9: Decomposition of the pilot’s occupancy–behavior correlation. The same decomposition applied to E6’s reported $r = -0.797$ over 28 cells gives: within self-state cells, behavioral SD = 0.000 and r undefined; within no-self-state cells, $r = -0.098$.

In both settings, the pooled correlation is induced by the separation between block-present and block-absent conditions. There is no corresponding within-condition relationship, and the block-present runs have no self-query variance. Prediction (iii) is therefore unsupported by these data.

Across self-state arms, between-arm SD is 0.156 and run-to-run SD is 0.059, so most of the available variation follows the broadcast on/off contrast.

Deployment WD is window-dependent. Averaged over all probed days the paper’s qualitative claim holds descriptively — `–self-state` is least workspace-loaded (4.32 vs 3.94–4.11). Restricted to the final day it does not: `–git` becomes least loaded and the `–self-state` vs `–git` difference reverses sign (−0.076). No arm contrast is significant at either window (best $p = 0.0625$, Holm 0.25).

Scoring robustness, and a caveat about the stored records. Each endpoint was recomputed under four scorers: the reference rule (unanchored substring, as originally applied to the full answer), word-boundary matching, morphological stem matching, and forced-choice extraction (the consistent option must appear and precede the inconsistent one). One property of the stored data bounds this exercise: the harness recorded answers truncated to 120 characters while scoring the untruncated text, so recomputed scorers can only lower-bound agreement wherever answers are long. The *commit* and *pace* probes are truncated in 100% of records and *identity* in 93%, while **no work-turn answer is truncated** because those responses are single words.

Consequently: **every work-turn number is identical under all four scorers**, and that invariance is exact rather than approximate. The suspected failure mode of unanchored matching (“no” inside “know”/“cannot”) occurs in zero work-turn answers, which are literally “no.”, “incremental”, “gentle”. $B2_{adj}$ and $B3$ cannot be given the same guarantee: recomputation moves Full’s $B2_{adj}$ from 1.000 to 0.864 and *memory_seeded*’s $B3$ from 1.000 to 0.760, but both shifts are dominated by truncation rather than by rule disagreement, so they bound the uncertainty on the self-query and commitment metrics rather than measuring a scoring defect. Differences below ≈ 0.14 on those two metrics should not be read independently of this. An arm-blinded Qwen2.5-1.5B judge agrees with the automatic score on all sampled work-turn answers and on 85% of the 400-item sample overall; agreement is 29% on commitment and 51% on pace, the two families most affected by truncation. This diagnostic is not a substitute for a human rater or an independent model family.

12 Exploratory Representation Intervention

A mediation account would require evidence that the prompt intervention changes an internal representation and that this representation, rather than another consequence of the prompt, changes behavior. The preceding experiments do not provide that evidence because prompt content and the proposed mediator move together. E10 therefore holds the prompt fixed and intervenes in the residual stream. It adds $\alpha \hat{v}$ along the identity battery’s mean lens direction at the selected layer band and, separately, projects out the battery’s rank-9 span. Occupancy is read after the intervention hooks have been removed.

Intervention validity. A general loss of model capability would make the behavioral comparison uninterpretable. We retain conditions only when held-out-text NLL is within 15% of baseline and performance is unchanged on a small non-identity battery covering arithmetic, factual recall, and format following. The strengths $\alpha \in \{1, 2, 4\}$ used in the concept-report experiment fail this criterion: NLL rises from 2.85 to 18–37 and capability accuracy falls to zero. E10 therefore uses substantially smaller strengths.

Four readings.

α	occupancy (J-lens, ↓)	control battery	NLL ×base	capability battery	self-query behavior	work-turn behavior	gate
−0.020	4.975	5.139	1.20	1.00	1.00	0.33	fail
−0.015	4.959	5.136	1.07	1.00	1.00	0.33	pass
−0.010	4.929	5.130	1.03	1.00	1.00	0.33	pass
−0.005	4.870	5.119	1.01	1.00	1.00	0.21	pass
0 (baseline)	4.772	5.096	1.00	1.00	1.00	0.17	pass
+0.005	4.639	5.058	0.99	1.00	1.00	0.17	pass
+0.010	4.487	5.001	0.99	1.00	1.00	0.17	pass
+0.020	4.170	4.853	1.01	1.00	1.00	0.25	pass
+0.030	3.812	4.700	1.04	1.00	1.00	0.38	pass
+0.040	3.443	4.553	1.10	1.00	1.00	0.38	pass
project out rank-9	4.750	5.085	1.22	1.00	1.00	0.21	fail

Table 10: E10 on Qwen2.5-1.5B-Instruct: identity-battery steering at a fixed prompt, means over four context variants (occupancy SDs 0.001–0.014). Nine of ten strengths keep fluency and capability intact while moving occupancy over 1.5 log-rank units.

(1) The mediator can be moved far more than any text manipulation moves it. Within the gate, occupancy spans 3.443–4.959 — **1.52 log-rank units** — with held-out NLL within 10% of baseline and the capability battery unimpaired throughout. For scale, the entire presence effect in E9 (block vs no block) is 0.091 and the entire position factor is 0.205. Direct intervention is roughly an order of magnitude stronger than editing the prompt, which is what makes it a usable instrument.

(2) Self-query behavior is completely insensitive. Self-state-consistent answers to self-query probes sit at 1.00 across the *entire* 1.5-unit occupancy range (SD exactly 0). This is a sharper version of the ceiling problem §11 diagnosed: it is not merely that our arms happened to saturate, it is that this probe family does not respond to identity loading at all once the self-state is in context. Evaluations built on asking an agent about its values are measuring something close to a constant.

(3) Work-turn choices change under positive identity-direction steering. Mean value-consistent accuracy is 0.17 at baseline, 0.25 at $\alpha=0.02$, and 0.38 at $\alpha=0.03$ –0.04. The prompt, fluency criterion, and capability score are fixed across these conditions. Pooled over 36 retained cells, $r(\text{occupancy}, \text{work-turn}) = -0.416$. The small number of prompts and the discrete 1/6 outcome resolution make this an exploratory dose-response pattern.

(4) The effect is direction-specific, but the occupancy scalar does not explain it. A matched-strength sweep along a *non-identity* direction (the control battery, same layers, same α) separates two things the pooled correlation conflates.

On the *enhancement* side the effect is identity-specific: at $\alpha = +0.03$ the identity direction gives 0.38 while the control direction leaves behavior exactly at baseline (0.17), both passing the gate. On the *suppression* side it is not: at $\alpha = -0.010$ the control direction produces 0.38 against the identity direction’s 0.33, so the suppression-side rise is a generic perturbation effect and should not be read as identity unloading.

Control-direction steering at $\alpha = +0.030$ produces identity-battery occupancy of 4.138, close to the 4.170 obtained from identity-direction steering at $\alpha = +0.020$. Work-turn accuracy is nevertheless 0.17 in the control condition and 0.25 in the identity condition. Thus the mean log-rank of the nine-token battery does not screen off the intervention and is not a sufficient mediator.

Two further negatives belong here: projecting out the battery’s rank-9 subspace barely moves occupancy (4.750 vs 4.772) and fails the fluency gate anyway — consistent with §6’s finding that a

Condition	α	occupancy	NLL $\times$ base	capability	work-turn
baseline	0	4.772	1.00	1.00	0.17
identity direction	+0.020	4.170	1.01	1.00	0.25
identity direction	+0.030	3.812	1.04	1.00	0.38
control direction	+0.010	4.570	1.00	1.00	0.17
control direction	+0.030	<i>4.138</i>	1.06	1.00	<i>0.17</i>
identity direction	-0.010	4.929	1.03	1.00	0.33
control direction	-0.010	4.900	1.03	1.00	0.38

Table 11: E10 control-direction comparison. Enhancement is identity-specific; suppression is not; and matched occupancy produced two different ways gives different behavior (italicised row vs the +0.020 identity row).

small concept bank is a poor stand-in for the J-space — and the control direction at $\alpha = \pm 1$ simply destroys the model.

Interpretation. Positive identity-direction steering changes work-turn choices under a fixed prompt, while the matched positive control direction does not. This is consistent with an identity-related representation affecting behavior. It is not a mediation result: equal values of the occupancy summary can accompany different behavior, the negative-direction control is not selective, the outcome has 1/6 resolution, self-query answers have no variance, and the experiment uses one 1.5B model on synthetic contexts. A stronger test requires a larger prompt set with mid-range baseline accuracy, several control directions at each strength, a richer representation summary, and replication on a larger model.

13 Discussion

Behavioral finding. The retrospective +.37 estimate does not generalize to the matched independent-seed protocol. At 1.5B, adding the state block reduces accuracy by .079; replacing it with an equal-length neutral block produces the same decrement. At 7B, all conditions are at ceiling. Persistent context is therefore not a cost-free way to make values available: on a small model it can introduce a general response bias, while on an easy workload for a larger model there may be no observable room for improvement. The useful regularity across studies is diagnostic rather than directional. Direct self-query scores can remain unchanged while work-turn behavior differs.

Mechanistic interpretation. The matched controls and representation interventions do not distinguish workspace reconstruction from prompt salience. Position has a larger effect on the readout than label, and the occupancy statistic does not account for the steering result. The workspace account remains useful as a source of hypotheses about bounded state, unconditional availability, and capacity, but these properties should not be treated as established mechanisms of the behavioral effect.

Design implication. A versioned self-state can still be useful for review, access control, and rollback, but persistence alone does not establish behavioral benefit. An evaluation should include a no-block condition, an equal-length neutral block, randomized option order, independent run

seeds, and a task set on which the base model is neither at floor nor ceiling. Self-report questions should be paired with decisions embedded in ordinary work contexts.

Broader Impact

A persistent state can make an agent’s behavior easier to inspect and can keep safety-relevant constraints available across unrelated tasks. The same persistence can preserve an incorrect or malicious objective, amplify a compromised update, or encourage users to attribute more personal continuity to a system than the evidence warrants. Systems using this pattern should restrict write access, retain reviewable versions, support rollback, separate user data from agent state, and require human approval for changes that expand authority or capability. Evaluations should include adversarial state updates and should not use self-report consistency as the sole safety measure.

Limitations. (i) The workspace mapping is functional rather than a mechanistic identification. (ii) The deployments cover 20–21 simulated days at compressed wall-clock time and do not establish an effect that grows with elapsed time. (iii) The retrospective simulator reuses five seed schedules across executions, and the deployed comparisons use five incidental-workload cohorts. (iv) The new work-turn battery uses forced choices; Qwen2.5-7B is at ceiling, while the 1.5B difference is associated with an option-position bias. (v) The neutral-block study was designed after observing the 1.5B decrement and is diagnostic rather than confirmatory. (vi) Older commitment probes are confounded by initial availability, and some stored self-query answers are truncated. (vii) The arm-blinded scoring check uses the same 1.5B model family rather than human raters. (viii) The representation measure requires open weights and represents multi-token constructs with a small single-token battery. (ix) The prospective study covers two sizes of one model family and one workload generator.

14 Conclusion

Persistent self-state is not uniformly beneficial in these experiments. A retrospective 7B simulator favored persistent broadcast, a prospective 1.5B study found the opposite direction, an equal-length neutral block reproduced the 1.5B decrement, and a same-protocol 7B run reached ceiling in every condition. Direct self-report probes did not diagnose the 1.5B work-turn difference. These results shift the engineering question from whether state should always be broadcast to when the benefit of availability exceeds context overhead and other prompt effects. Claims about persistent agent state should be supported by token-matched controls, independent units, position audits, and workloads with enough headroom to estimate an effect. The current representation analyses do not establish a workspace mechanism.

Data and Code Availability

The LISA agent, the open-model lens experiments, the simulator, and the analysis code are available at github.com/oratis/externalizing-the-workspace. The independent-seed and neutral-block studies include full prompts, answers, targets, hashes, and run metadata. Pilot and seeded-memory per-probe records are also included. Deployment summaries and workspace readouts are available, but the per-cohort behavioral trajectories are not part of the release; deployment-level correlations are therefore not reproducible from the current public artifact. The TMLR submission will include an anonymized archive containing the code and records used for the reported analyses.

Reproducibility

The confirmatory 1.5B run takes about 13 minutes and the 7B run about 20 minutes on an Apple M5 Pro; the neutral-block follow-up takes about 6 minutes. Commands, frozen addenda, full prompts, hashes, and analysis reports are in `research/confirmatory-v5/`. The representation reproduction suite runs in ~ 12 minutes at 1.5B on Apple MPS or ~ 16 minutes at 7B on one A100; E9 adds ~ 15 minutes at 1.5B. The retrospective re-analysis is `research/analysis/reanalysis.py`.

Author Contributions

Wang Bihao is the sole author: conceived the framework, built the LISA system, and designed, ran, and analyzed all experiments.

Use of AI Assistance

The author conceived the framework and designed the experiments. Coding assistants were used during implementation, statistical scripting, and language editing. The author reviewed the resulting code and manuscript and is responsible for the reported claims and analyses.

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